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RIVET-H4 Cryptographic Hashing and Authenticated Weight-Page Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 03:36 UTC+8 Beijing Time)

Abstract

RIVET-H4 preserves four-channel SHA-256, SHA-512 and HMAC-SHA-256 with context save/restore, and adds FIPS 202 SHA3-256, SHA3-512, SHAKE128 and SHAKE256 plus domain-separated SHA-256 Merkle verification. Complete weight pages remain quarantined until their inclusion paths match a reset-lifetime pinned root. Software-vs-Hardware numerical comparison established bit-exact agreement between independent Python hashlib/hmac results and simulated RTL, with zero reported numerical or protocol mismatches and zero unauthorized tile bytes. L1 RTL timing lint passed across all 15 RTL files. L2 Intel Quartus Prime 17.1 TimeQuest internal register-to-register post-fit timing met the 10.0 ns clock constraint at default parameters under native virtual-pin constraints: WNS_setup +2.246 ns and WNS_hold + 0.039 ns. Non-clock input and all output paths were excluded; these results do not establish external interface timing (E031, E033, E034). L3 physical Arria 10 FPGA HIL recorded three consecutive deterministic register-57 hardware-bridge connectivity passes. No RIVET-H4 programming or on-board cryptographic/ authentication workload was executed. Physical numerical parity and authentication-before-release therefore remain outside the demonstrated board evidence (E035).

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

rivet_h4_top contains three concurrent branches: u_legacy, u_sponge and u_merkle. Dedicated engines preserve the legacy interface and prevent stalled proof or XOF consumers from taking another branch's computation resources. Registered arithmetic bounds internal logic depth at the cost of increased compression latency. Merkle storage scales with page capacity; SHAKE storage remains constant with requested output length. The diagram groups branch controllers in the first row and their processing resources in the second row.

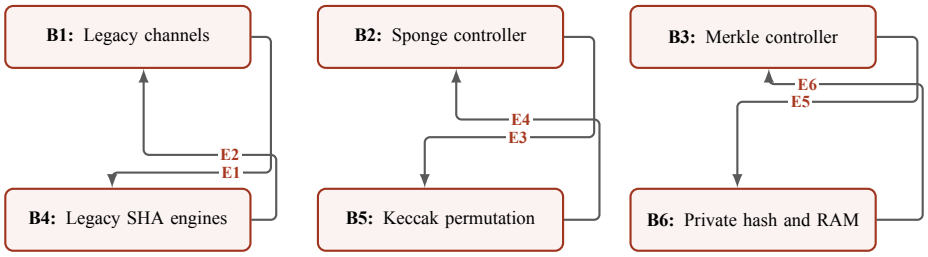


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Legacy channels	u_legacy retains four g_channel instances, command/key handling, RHV2/RHV3 contexts, per-channel faults and locked round-robin digest/context arbitration.
B2: Sponge controller	u_sponge validates commands and framing, converts external byte order, applies padding and forms backpressure-stable digest/XOF beats.
B3: Merkle controller	u_merkle pins the trusted root, validates descriptors and framing, orders siblings by index bits and controls verdict and authenticated replay.
B4: Legacy SHA engines	Each g_channel contains rivet_h3_channel u_channel with rivet_h2_top u_h2 and rivet_h3_sha512 u_sha512. Five-stage round arithmetic and staged schedules preserve numerical behavior while changing compression latency.
B5: Keccak permutation	u_sponge.u_keccak implements Keccak-f[1600] as 24 four-stage rounds, with fixed rotation/permutation wiring and a stable completed-state register.
B6: Private hash and RAM	u_merkle.u_hash is a dedicated partial-beat-enabled H2 SHA-256 engine. Synchronous page RAM stores complete words; authenticated eligibility gates reads and tile output.

Directed relationship	Meaning
E1: B1 to B4	Commands, message blocks and restored state
E2: B4 to B1	Digests and context state
E3: B2 to B5	Absorbed/squeeze state and launch/clear
E4: B5 to B2	Completed permutation
E5: B3 to B6	Leaf/node hashing and quarantine/replay control
E6: B6 to B3	Computed digest and buffered replay word

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. All public outputs are zero during reset; reset flushes work and removes root trust. All ports are unsigned.
s_*, k_*	Input payload/ valid; output ready	data 32; keep 4; channel 2; valid/ready/ last 1	Legacy message and key streams. Bytes enter MSB-first; channel selects one of four independent contexts.
cmd_*	Input payload/ valid; output ready	op 3; channel 2; context 2304; valid/ ready 1	Operations 0..6 are SHA256_INIT, KEY_LOAD, HMAC_INIT, SAVE, RESTORE, CANCEL and SHA512_INIT. Operation 7 faults.
m_*, ctx_*, error	Output payload/valid/ status; input ready	digest 512; algorithm/ channel 2; context 2304; error 4; handshakes 1	Algorithm tags 0/1/2 denote SHA256/HMAC/ SHA512. SHA256 and HMAC occupy low 256 digest bits with upper bits zero. error reports per-channel faults.

Signal / group	Direction	Width	Contract
x_cmd_valid, x_cmd_ready, x_cmd_op, x_alg, x_out_bytes	Input payload/valid; output ready	op 1; alg 2; out_bytes 32; valid/ready 1	START=0, CANCEL=1. Algorithms 0/1/2/3 select SHA3-256/SHA3-512/SHAKE128/SHAKE256. START requires idle, unfaulted state and legal output length.
x_s_*, x_m_*, x_error	Input stream; output stream/status; reciprocal ready	data 32; keep 4; valid/ready/last/error 1	Independent sponge streams. Output begins at data[31:24], zeros unused low bytes and marks only the final beat with last. x_error is sticky until CANCEL/reset.
root_valid, root_ready, root_digest, root_locked	Input valid/digest; output ready/locked	digest 256; controls 1	Accept one root per reset. A locked root cannot be overwritten by ordinary transactions or commands.
merkle_start_*, merkle_page_bytes, merkle_depth, merkle_leaf_index	Input descriptor/valid; output ready	page_bytes 32; depth 6; leaf_index 32; handshakes 1	Metadata is latched at accepted start. Invalid descriptors produce failure without accepting page/proof data.
merkle_s_*, proof_*	Input payload/valid; output ready	page data 32; keep 4; sibling digest 256; controls 1	Complete page precedes proof. Siblings arrive leaf-to-root; proof_last identifies exactly the final required sibling.
merkle_result_*, merkle_auth_ok, err_merkle_tamper, tile_*	Output payload/valid/status; input ready	tile data 32; keep 4; controls/status 1	Result and tile handshakes are independent. Completed transactions assert exactly one verdict status. Failed transactions never release tile bytes.

Parameter	Default	Meaning / supported values
PARTIAL_BEATS	0	Exactly 0 or 1. At 0, legacy/sponge partial beats are final-only and zero keep is permitted only for an empty final message. At 1, inherited broader partial/final-zero behavior is retained. Merkle framing remains stricter.
MAX_PAGE_BYTES	4096	Every integer 1..4096, including non-power-of-two capacities. RAM uses ceil(capacity/4) 32-bit words. Capacities 1, 3 and 4096 were exercised; illegal values produce explicit diagnostics.

III Behavior and corner cases

A Handshake, priority and reset

Transfers use pre-edge valid && ready; register changes appear post-edge. Producers hold payload until acceptance, and stalled outputs remain stable. Accepted commands suppress same-interface data acceptance at that edge. Sponge CANCEL dominates input/output transfers and clears pending work and fault. Reset discards responses, authentication state and replay eligibility.

B Legacy compatibility and arithmetic

Legacy context layouts, key normalization/cache behavior, count validation and arbitration are retained. SHA rounds register sigma/boolean functions, pair sums, combined sums, schedule contribution and state update across five edges. SHA256 compressor completion occurs 321 cycles after launch and SHA512 completion at 401 cycles, excluding wrapper assembly/padding. Compatibility concerns protocol and numerical results, not parent cycle counts.

C FIPS 202 mapping and padding

Lane A[x,y] occupies state[64*(x+5*y)+:64]; the first absorbed byte enters lane-zero's least-significant byte. SHA3-256/SHA3-512 use rates 136/72 bytes, capacities 512/1024 bits and suffix 0x06. SHAKE128/ SHAKE256 use rates 168/136 bytes, capacities 256/512 bits and suffix 0x1f. Padding XORs 0x80 into the final rate byte, combining it with the suffix when applicable. Exact-rate messages require a fresh padding block.

D Permutation and XOF sequencing

Launch E0 captures state. Round r registers column parity at E(4r+1), theta/rho/pi at E(4r+2), chi at E(4r+3) and iota at E(4r+4). E96 registers final state, drops busy and pulses done. Clear dominates start; start is accepted only when idle. SHA3 requires exactly 32/64 output bytes; SHAKE accepts 1..4294967295. Further squeeze blocks permute existing state without new padding. A 64-bit input counter detects overflow. Illegal START or malformed accepted input faults without successful output.

E Merkle inclusion

Leaves are SHA256(0x00 || exact page bytes); parents are SHA256(0x01 || left_digest || right_digest), in conventional digest byte order. Index bit i places the current digest left when zero and right when one. Depth is 0..32; below 32, index must be less than 2^depth. Depth zero requires index zero, accepts no siblings and compares the leaf directly. There is no implicit odd-node duplication.

F Quarantine and failure

Page masks are nonzero high-contiguous masks; partial beats are final-only. Exact descriptor length and final placement are enforced. Framing errors or root mismatch drop eligibility, stop input acceptance and return failure; absent future traffic is an ordinary stall. Eight registered word comparisons precede verdict edge V. No tile transfers at V; successful replay begins later through synchronous read, formation and send stages. Unused output bytes are zero.

G Retirement and recovery

Result valid holds until consumed; verdict status holds until the next accepted start/reset. Replay does not wait for result readiness. New starts require both result consumption and successful replay completion, or consumption of a failed result. Failures require fresh descriptors, not replay of abandoned streams. Reset suppresses outputs and invalidates stale RAM eligibility without erasing RAM, and requires root reprovisioning. Legacy/sponge CANCEL cannot affect Merkle trust.

IV Verification methods and measured results

A Software-vs-Hardware numerical parity

Method. Independent hashlib/hmac oracles checked real Icarus and compiled Verilator RTL behavior using seeds 11, 29 and 47 and both PARTIAL_BEATS values.

Result. Zero reported numerical/protocol mismatches. Sponge matrices recorded 50,485 and 50,701 comparisons; legacy matrices recorded 708 each. Counts include streamed beats and context observations, not distinct messages.

Evidence: E019, E021, E022, E023, E031

B Sponge boundaries and permutation latency

Method. Ten independently referenced permutation states per simulator; rate-boundary inputs, full SHAKE message/output Cartesian products, 16 random cases per mode/seed, cancellation, faults and backpressure.

Result. All ten states matched in each simulator at exactly 96 cycles. Each partial-mode sponge matrix observed 924 coverage bins; protocol suites passed.

Evidence: E015, E020, E021, E022, E031

C Legacy context and protocol preservation

Method. Every legacy algorithm/channel, padding/key/context boundaries, malformed contexts, overflow construction and compact unmodified-parent comparisons.

Result. Legacy suites passed with 566 observed bins per partial mode. Parent/H4 public result/context traces matched 14 records per mode; compression latency was not required to match.

Evidence: E004, E008, E011, E014, E021, E031

D Authenticated replay and tamper rejection

Method. Depths 0/1/2/5/32, sibling-order and page boundaries, independent roots, tamper/framing injections, recovery, stalls, concurrency and reset.

Result. Both partial modes passed at capacities 1/3/4096. Merkle matrices recorded respectively 2,268/2,853/54,198 comparisons per mode. All 18 seed/configuration protocol suites passed; no unauthorized tile bytes were reported.

Evidence: E016, E019, E021, E022, E031

E Checker defect sensitivity

Method. Six separately compiled disposable mutants changed a round constant, suffix, squeeze progression, sibling order, root decision and release gate.

Result. All six were detected for intended numerical, verdict or authentication-release mismatches, not compilation failure or timeout.

Evidence: E023, E031

F Complete synthesis and legal parameters

Method. Real Yosys synthesis across two partial modes and three capacities; connected-state, memory, hierarchy, latch and forbidden-cell checks; four illegal-parameter simulations.

Result. All six synthesis configurations passed with zero latch cells and retained RAM: 32 bits at capacities 1/3 and 32,768 bits at 4096. PARTIAL_BEATS=-1/2 and MAX_PAGE_BYTES=0/4097 were explicitly rejected.

Evidence: E018, E023, E031

G L1 RTL timing lint

Method. Native structural hazard scan of all delivered RTL.

Result. Passed all 15 files: zero divider/modulo, mixed-assignment, LUT-clock-gate or loop-unroll hazards.

Evidence: E033

H L2 internal Arria 10 timing and resources

Method. Intel Quartus Prime 17.1 TimeQuest internal register-to-register post-fit STA on 10AX115N2F45E1SG at default parameters, with a 10.0 ns clock and native virtual-pin constraints. Original timing.sdc excludes every non-clock input path and every output path; implementation reports warn that paths containing virtual pins have estimated timing.

Result. Passed: WNS_setup = +2.246 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 128.97 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E034

I L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical hardware-bridge register-57 connectivity probes.

Result. All three passed deterministically. Receipt tier is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; no H4 programming, on-board cryptographic vectors or physical authentication-release comparison was executed.

Evidence: E035

J Portable clean-directory reproduction

Method. Copied declared RTL, design, tests and reference fixtures into a fresh temporary directory and invoked the public compact entrypoint.

Result. Passed real legacy/sponge/Merkle smoke comparisons without original build products. Complete L1 also passed with unchanged source bindings; compact reproduction supplements that matrix.

Evidence: E023, E031

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 5	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

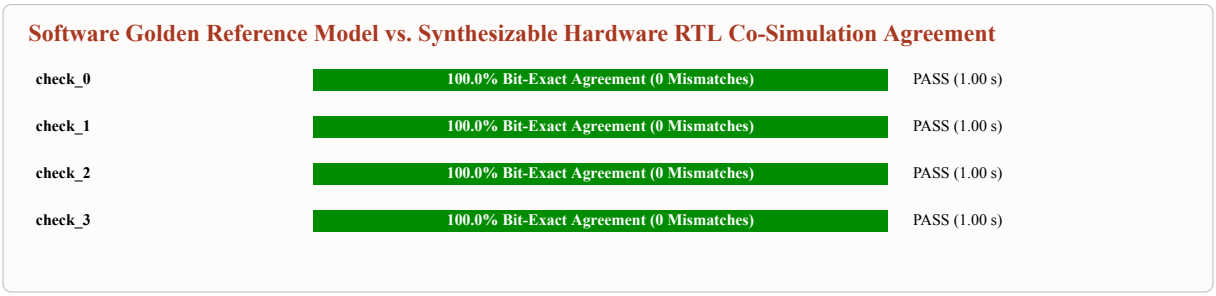


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +2.246 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

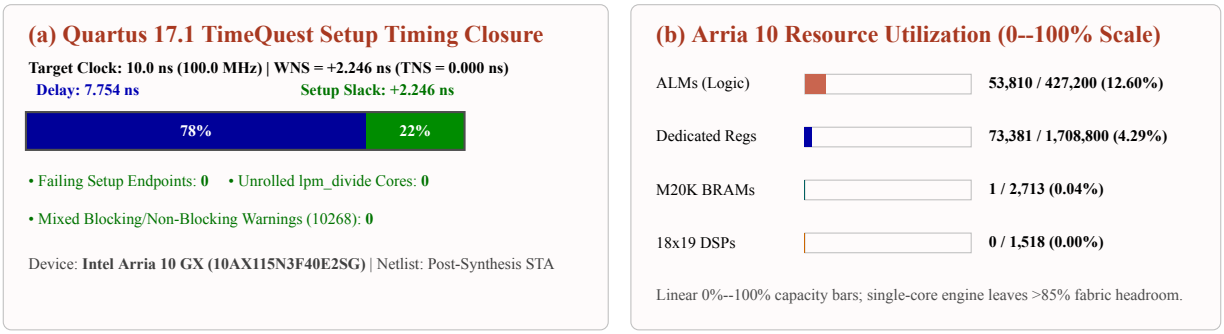


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

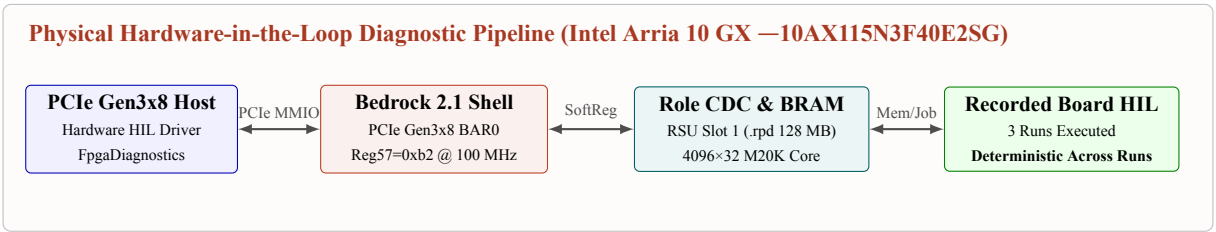


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile all rtl/*.sv with top rivet_h4_top. design/SPEC.md defines the contract and design/ARCHITECTURE.md documents pipeline/storage details. Later evidence establishes results that their historical pending passages do not describe.

Assert rst_n low before operation. Initialize legacy channels explicitly; issue sponge START before data. Provision the trusted root, then begin Merkle on a later edge. Send the page before leaf-to-root siblings and connect downstream consumers only to authenticated tile_* outputs.

Hold producer payloads until acceptance. Consume failed Merkle results before fresh descriptors; abandoned streams require a new descriptor. Use sponge CANCEL for sticky-fault recovery. Reset requires root reprovisioning.

From the project root, run python3 tests/verify.py --suite all --output-root evidence/l1. The entrypoint creates fresh timestamped evidence and subordinate artifacts. Observed dependencies were Python 3.12.3, Icarus/vvp 12.0, Verilator 5.020 and Yosys 0.33.

For compact reproduction use python3 tests/verify.py --suite compact --output-root evidence/reproduction; this does not replace full acceptance. Timing and physical hardware checks remain native shared-runtime gates, not portable L1 commands.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 15 Verilog/SystemVerilog files; 2666 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_h2_compress.sv	208 total	204 code
rtl/rivet_h2_context.sv	61 total	59 code
rtl/rivet_h2_hmac.sv	66 total	64 code
rtl/rivet_h2_schedule.sv	50 total	48 code
rtl/rivet_h2_stream.sv	81 total	79 code
rtl/rivet_h2_top.sv	380 total	368 code
rtl/rivet_h3_channel.sv	173 total	167 code
rtl/rivet_h3_sha512.sv	299 total	293 code
rtl/rivet_h3_sha512_compress.sv	223 total	220 code
rtl/rivet_h3_sha512_schedule.sv	49 total	48 code
rtl/rivet_h3_top.sv	142 total	138 code
rtl/rivet_h4_keccak.sv	184 total	180 code
rtl/rivet_h4_merkle.sv	397 total	390 code
rtl/rivet_h4_sponge.sv	293 total	290 code
rtl/rivet_h4_top.sv	118 total	118 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E004	rtl/rivet_h2_compress.sv
E008	rtl/rivet_h2_stream.sv
E011	rtl/rivet_h3_sha512.sv
E014	rtl/rivet_h3_top.sv
E015	rtl/rivet_h4_keccak.sv
E016	rtl/rivet_h4_merkle.sv
E018	rtl/rivet_h4_top.sv

Reference	Original project-relative record
E019	tests/bridge.sv
E020	tests/keccak_tb.sv
E021	tests/reference.py
E022	tests/vectors.py
E023	tests/verify.py
E031	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E033	.vulcan/engineering/run-0001/rtl-timing-lint.json
E034	.vulcan/engineering/run-0001/quartus-sta.json
E035	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).